



SEA-BIRD
SCIENTIFIC

SBE37-SMP-ODO MicroCAT

Instrument Configuration

Instrument Serial Number: 37-16244
Instrument Firmware Version: 2.13.0
Zero Conductivity Frequency: 2805.10
Communications Format: RS232
Communications Settings: 9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

<i>Data Format</i>	<i>Measurement</i>	<i>Sensor Type</i>	<i>Serial Number</i>	<i>Rating</i>
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10746411	2000m(2000 dBar)
RS232	Oxygen	SBE 63	63-1754	7000m

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.



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SENSOR SERIAL NUMBER: 16244
CALIBRATION DATE: 03-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

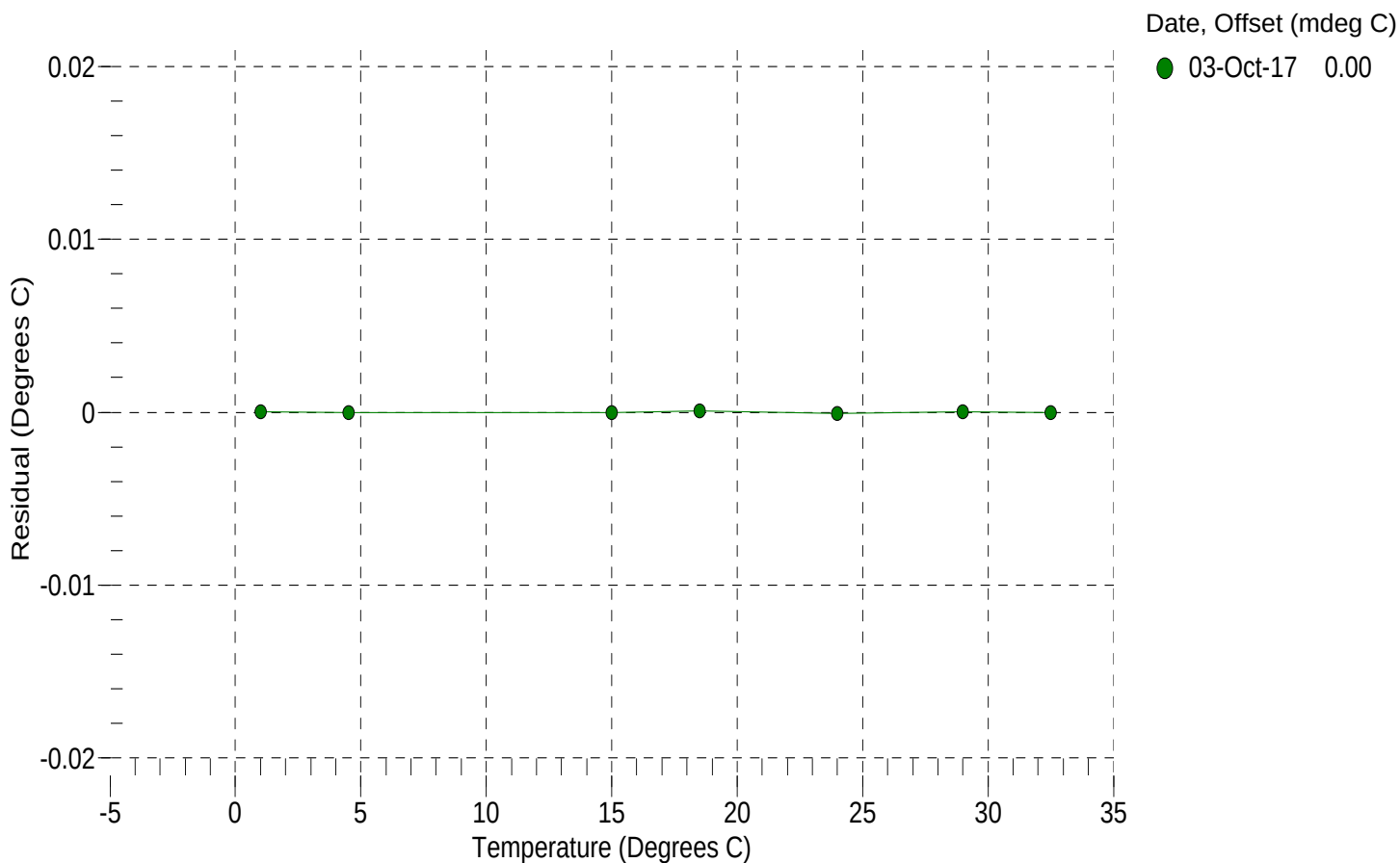
a0 = -1.751506e-004
a1 = 3.181547e-004
a2 = -5.149822e-006
a3 = 2.196750e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	568730.2	1.0000	0.0000
4.5000	487117.4	4.5000	-0.0000
15.0000	312094.0	15.0000	-0.0000
18.5000	270724.7	18.5001	0.0001
24.0000	217804.3	23.9999	-0.0001
29.0000	179816.2	29.0000	0.0000
32.5000	157763.0	32.5000	-0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1 / \{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature





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SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

g = -1.004872e+000
h = 1.277796e-001
i = -8.766068e-005
j = 2.198588e-005

CPcor = -9.5700e-008
CTcor = 3.2500e-006
WBOTC = 1.1882e-007

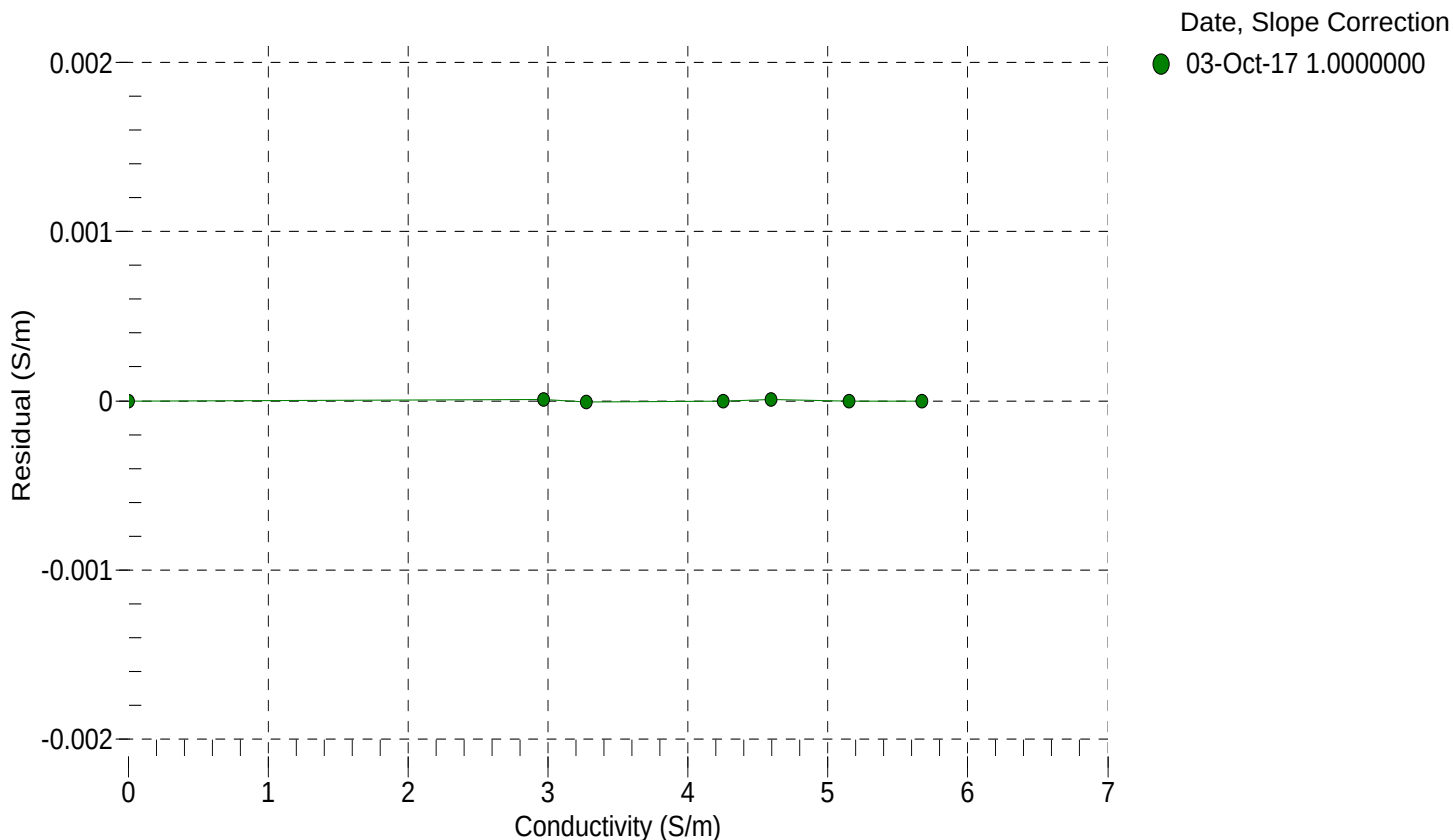
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2805.10	0.00000	0.00000
1.0000	34.6932	2.96644	5570.68	2.96645	0.00001
4.5000	34.6732	3.27255	5780.63	3.27254	-0.00001
15.0000	34.6306	4.25124	6405.18	4.25123	-0.00000
18.5000	34.6218	4.59534	6610.51	4.59535	0.00001
24.0000	34.6123	5.15162	6929.30	5.15162	-0.00000
29.0000	34.6073	5.67193	7214.43	5.67193	-0.00000
32.5000	34.6050	6.04330	7409.20	6.03974	-0.00356

$f = \text{Instrument Output(Hz)} * \sqrt{1.0 + \text{WBOTC} * t} / 1000.0$

t = temperature (°C); p = pressure (decibars); $\delta = \text{CTcor}$; $\epsilon = \text{CPcor}$;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \epsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity





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SENSOR SERIAL NUMBER: 16244
CALIBRATION DATE: 29-Sep-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10746411

COEFFICIENTS:

PA0 =	3.590349e-001	PTCA0 =	5.235554e+005
PA1 =	8.991203e-003	PTCA1 =	1.466240e+000
PA2 =	-1.426465e-010	PTCA2 =	1.223803e-002
PTEMPA0 =	-6.042631e+001	PTCB0 =	2.498300e+001
PTEMPA1 =	5.453992e-002	PTCB1 =	3.990025e-004
PTEMPA2 =	-6.552239e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

THERMAL CORRECTION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)	TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
14.59	525181.0	1561.8	14.61	0.00	32.50	1740	525282.35
591.72	589459.6	1564.9	591.71	-0.00	29.00	1673	525279.59
1168.84	653861.0	1565.7	1168.74	-0.00	24.00	1578	525270.68
1746.00	718418.3	1566.3	1745.98	-0.00	18.50	1473	525255.94
2323.13	783106.5	1566.9	2323.19	0.00	15.00	1407	525245.86
2899.55	847818.2	1567.3	2899.43	-0.00	4.50	1208	525231.06
2323.07	783111.9	1566.5	2323.24	0.01	1.00	1142	525227.36
1746.03	718422.8	1565.6	1746.02	-0.00	TEMPERATURE (°C) SPAN (mV)		
1168.85	653869.7	1564.8	1168.82	-0.00			
591.68	589456.5	1564.4	591.68	0.00			
14.59	525182.5	1564.1	14.62	0.00	-5.00		24.98
					35.10		25.00

y = thermistor output (counts)

t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y²

x = instrument output - PTCA0 - PTCA1 * t - PTCA2 * t²

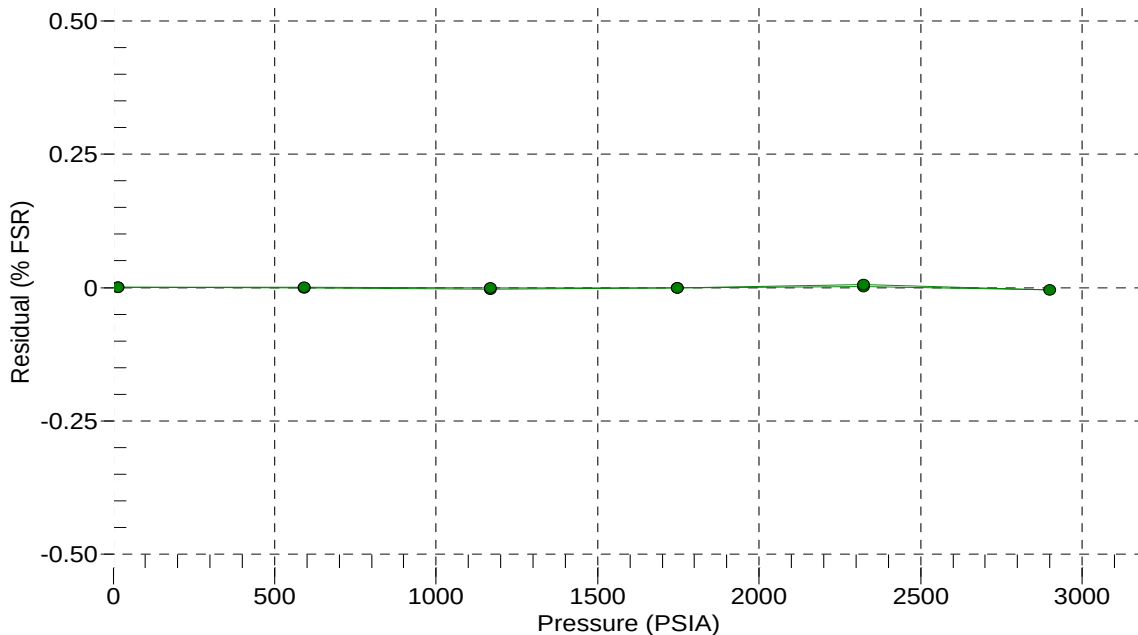
n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t²)

pressure (PSIA) = PA0 + PA1 * n + PA2 * n²

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range

Date, Offset (%FSR)

● 29-Sep-17 0.00





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Pressure Test Certificate

Test Date: **2017-09-28**

Description: **SBE-37 Microcat**

Sensor Information:

Model Number: **SBE-37**

Serial Number: **16244**

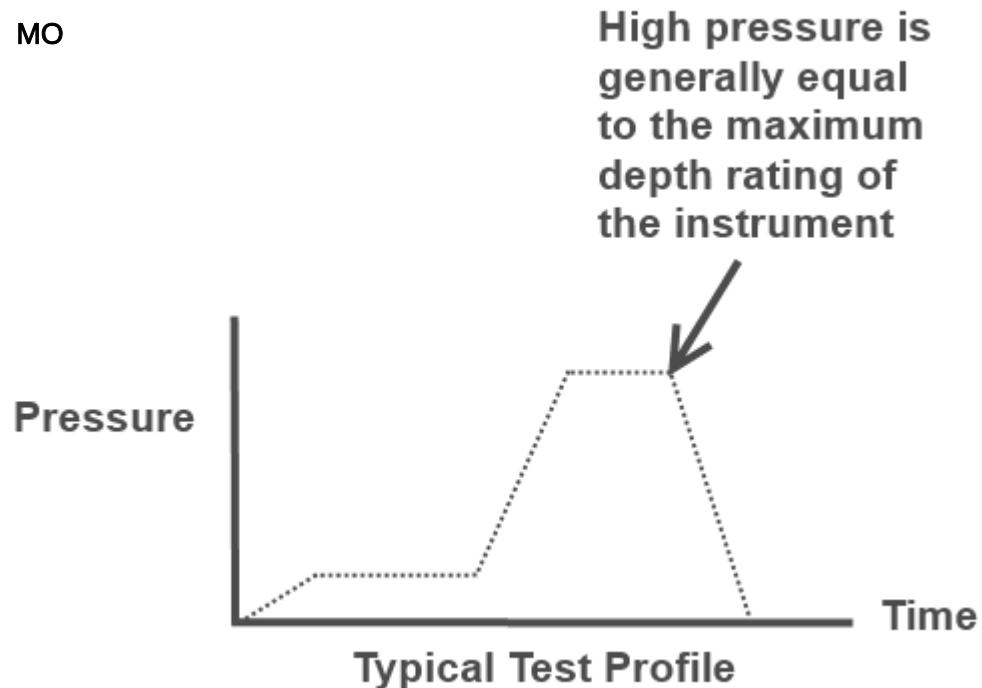
Pressure Test Protocol:

Low Pressure Test: **40** PSI Held For: **15** Minutes

High Pressure Test: **2900** PSI Held For: **15** Minutes

Passed Test: **True**

Tested By: **MO**





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SENSOR SERIAL NUMBER: 1754
CALIBRATION DATE: 20-Sep-17

SBE 63 OXYGEN TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

TA0 = 7.340040e-004 TA2 = 1.616127e-006

TA1 = 2.412989e-004 TA3 = 7.124288e-008

BATH TEMP (° C)	INSTRUMENT OUTPUT(V)	INST TEMP (° C)	RESIDUAL (° C)
1.9999	1.12482	1.9996	-0.00025
1.9999	1.12481	2.0000	0.00006
2.0000	1.12481	2.0000	-0.00004
2.0001	1.12480	2.0003	0.00016
6.0000	1.00022	6.0001	0.00007
6.0000	1.00022	6.0001	0.00007
6.0001	1.00022	6.0001	-0.00003
6.0001	1.00022	6.0001	-0.00003
12.0000	0.83376	12.0001	0.00010
12.0000	0.83376	12.0001	0.00010
12.0001	0.83376	12.0001	-0.00000
12.0001	0.83376	12.0001	-0.00000
19.9999	0.64929	19.9997	-0.00017
20.0000	0.64928	20.0002	0.00022
20.0001	0.64929	19.9997	-0.00037
20.0001	0.64929	19.9997	-0.00037
25.9999	0.53671	26.0005	0.00056
25.9999	0.53671	26.0005	0.00056
26.0000	0.53672	25.9999	-0.00013
26.0000	0.53672	25.9999	-0.00013
30.0000	0.47255	30.0000	0.00003
30.0000	0.47256	29.9994	-0.00063
30.0002	0.47255	30.0000	-0.00017
30.0003	0.47254	30.0007	0.00040

V = Instrument Output (Volts)

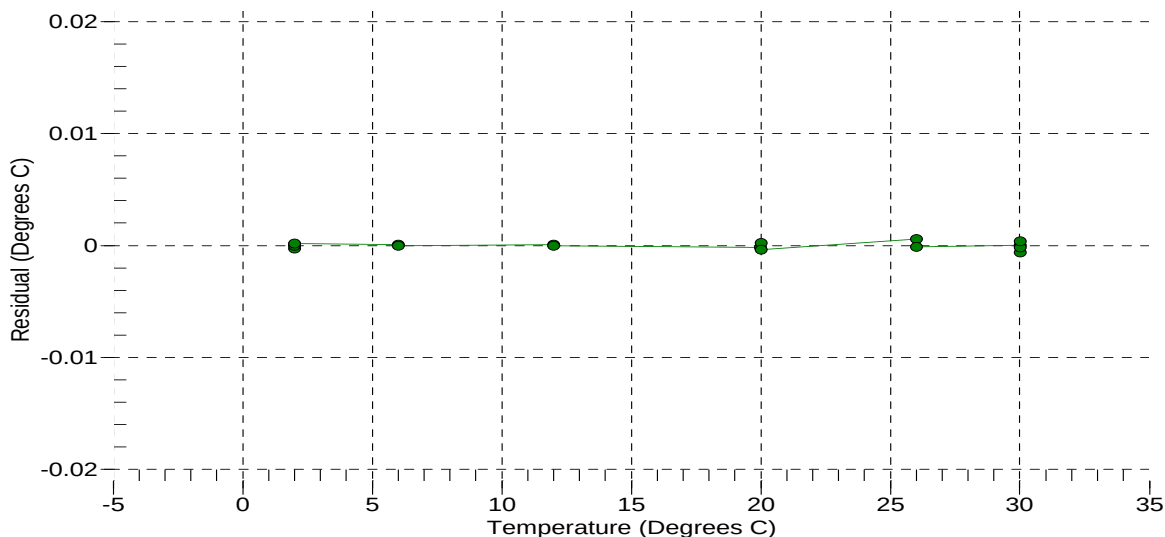
$L = \ln(100000 * V / (3.3 - V))$

Temperature ITS-90 (°C) = $1 / (TA0 + (TA1 * L) + (TA2 * L^2) + (TA3 * L^3)) - 273.15$

Residual (°C) = instrument temperature - bath temperature

Date, Offset (mdeg C)

● 20-Sep-17 -0.00





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SBE 63 OXYGEN CALIBRATION DATA

COEFFICIENTS:

A0 = 1.0513e+000 B0 = -2.4869e-001 C0 = 1.0464e-001 E = 1.1000e-002
A1 = -1.5000e-003 B1 = 1.6248e+000 C1 = 4.4532e-003
A2 = 3.9296e-001 C2 = 6.1229e-005

BATH OXYGEN (ml/l)	BATH TEMPERATURE (° C)	BATH SALINITY (PSU)	INSTRUMENT OUTPUT (µsec)	INSTRUMENT OXYGEN (ml/l)	RESIDUAL (ml/l)
0.714	30.00	0.00	31.18	0.712	-0.002
0.732	26.00	0.00	31.92	0.731	-0.001
0.788	20.00	0.00	32.83	0.789	0.001
0.874	12.00	0.00	34.21	0.872	-0.002
0.980	6.00	0.00	35.10	0.979	-0.001
1.077	2.00	0.00	35.65	1.078	0.001
2.190	30.00	0.00	22.87	2.190	-0.001
2.325	26.00	0.00	23.40	2.326	0.001
2.468	20.00	0.00	24.55	2.473	0.006
2.982	12.00	0.00	25.45	2.981	-0.002
3.396	6.00	0.00	26.39	3.395	-0.002
3.643	30.00	0.00	18.91	3.642	-0.001
3.749	2.00	0.00	27.03	3.746	-0.003
3.899	26.00	0.00	19.33	3.903	0.004
4.319	20.00	0.00	20.06	4.330	0.010
5.071	12.00	0.00	21.11	5.070	-0.002
5.194	30.00	0.00	16.38	5.187	-0.007
5.607	26.00	0.00	16.70	5.604	-0.003
5.815	6.00	0.00	21.95	5.814	-0.001
6.221	20.00	0.00	17.33	6.229	0.008
6.432	2.00	0.00	22.53	6.437	0.005
7.286	12.00	0.00	18.29	7.282	-0.004
8.344	6.00	0.00	19.07	8.339	-0.005
8.858	2.00	0.00	19.93	8.859	0.001

T = temperature (°C) , P = pressure (dbar), U = Instrument output (µsec)

S_{corr} (salinity correction function) = 1.0 for calibration in DI water

See the user manual for more information on S_{corr} calculation

$V = U / 39.457071$

Oxygen (ml/l) = $\{((A0 + A1 \cdot T + A2 \cdot V^2) / (B0 + B1 \cdot V) - 1.0) / (C0 + C1 \cdot T + C2 \cdot T^2)\} \cdot S_{corr} \cdot \exp(E \cdot P / (T + 273.15))$

Residual (ml/l) = instrument oxygen - bath oxygen

Date, Slope Correction

● 20-Sep-17 1.0000

